International Journal of Modern Physics B, Vol. 14, No. 9 (2000) 889–907
© World Scientific Publishing Company

TRANSITIONS BETWEEN STATIONARY STATES OF THE ELECTRON–PHOTON FIELD SYSTEM: A NEW MECHANISM FOR INTERPRETING HIGH OPTICAL FIELD PHENOMENA

J. GAO, F. SHEN and J. G. EDEN

University of Illinois, Department of Electrical and Computer Engineering, Urbana, IL 61801, USA

Received 20 March 2000

Optical transitions between stationary states of the electron–photon field system, known as quantum Volkov states, offer a new framework within which to interpret a rich variety of phenomena observed in intense optical fields. In this view, photons and electrons interact to form transient complexes which are capable of annihilating fundamental photons to produce more energetic radiation, or releasing photons back to the optical field. That is, they mediate the exchange of energy between the fundamental and secondary optical fields. The application of this new picture, which has no classical analog, to high harmonic generation is reviewed here. In previous publications [J. Gao, F. Shen, and J. G. Eden, *Phys. Rev. Lett.* **81**, 1833 (1998); *J. Phys.* **B32**, 4153 (1999); *Phys. Rev.* **A61**, 043812 (2000)], theoretical predictions of harmonic spectra and the polarization characteristics of individual harmonics have been compared with semiclassical calculations and experiment and several key results are summarized here. Specifically, an analytic expression for the harmonic generation transition matrix element has been derived that is valid for an arbitrary polarization of the fundamental optical field and is in agreement with the experimental data in the literature. Also, recent calculations of the spectral lineshapes of individual harmonics are presented here and compared with published measurements. QED theory predicts that the harmonic linewidth is essentially independent of the fundamental intensity when the ponderomotive potential is less than the atomic ionization potential but increases rapidly at higher intensities. The implications of these results for future experiments are also discussed.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

1. Introduction

The electron–photon interaction is one of fundamental importance in physics and has taken on renewed significance over the past decade as advances in laser technology have resulted in the ability to generate optical field intensities of unprecedented magnitude. The impact of the mode-locked Ti-doped sapphire ($Ti : Al_2O_3$) laser, in particular, on atomic and optical physics over the past several years rivals that of the tunable dye laser more than two decades ago. From its discovery in 1982, the $Ti : Al_2O_3$ laser has been developed rapidly and, when mode-locked, is now capable of generating sub-20 fs pulses having energies beyond 1 mJ at kHz pulse repetition frequencies (PRF). Consequently, this laser offers access to a temporal and intensity

realm that promises to be as fruitful as was the dye laser in spectroscopy in the frequency domain.

As peak intensities have progressed past 10^{12} W·cm^{-2} to beyond 10^{18} W·cm^{-2}, experimentalists have observed a remarkable variety of nonlinear phenomena that are attributable to the optical field strengths that can now be generated. High order harmonic generation (HG), for example, is the term given to the production of odd harmonics of the fundamental when an intense optical field interacts with an atom or molecule in the gas phase.[1–3] Harmonics extending into the X-ray region have been generated[4] with efficiencies sufficient to warrant the expectation of table top X-ray sources in the near future having milliwatt average powers. Currently, approximately 1 nJ/pulse at a PRF of 1 kHz has been generated for individual harmonics. A similar effect is observed when an intense, ultrafast laser pulse impinges on a solid target except that both odd and even harmonics of the fundamental are produced.[5]

In addition to the observation of HG, experimentalists have studied the impact of an intense, external optical field on several well-known atomic and molecular photoprocesses. Consider, for example, e$^-$-atom scattering, Auger decay, and the photoelectric effect (photoionization). In the presence of an intense, external optical field, the electron energy spectra for each process exhibits new features, "sidebands" that are separated from the zero-field peak(s) by multiples of the external (imposed) photon energy.[6–8] The latter two effects have been termed laser-assisted Auger decay (LAAD)[7] and the laser-assisted photoelectric effect (LAPE).[8] All of these phenomena have a common origin. That is, HG, LAAD, and LAPE are different manifestations of the same interaction — that between an electron and an intense optical field. It is the thesis of this paper that this interaction is mediated by quantum Volkov states, which are stationary solutions to Schrödinger's (time-independent) equation for an electron interacting with a quantized optical field. Here, we will focus on HG because it is this process that originally motivated our theoretical efforts. Although the remainder of this paper will discuss the mathematical basis and physical interpretation of a new picture for the electron–photon interaction, theoretical predictions will be presented here only for HG. The implications of this framework for LAPE, LAAD, and other phenomena will be described elsewhere.

2. Theory of Transitions Between Quantum Volkov States

2.1. *Semiclassical models of harmonic generation*

Harmonic generation experimental results have, to date, almost exclusively been interpreted within a classical or semiclassical framework. The most prevalent picture is that of the two step model[9,10] in which a bound electron tunnels into the continuum under the influence of the intense, external field. In this framework, the optical field serves only to suppress the Coulombic barrier and accelerate (or decelerate) the electron. The interaction of the electron with the optical field is treated

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

classically and radiation is produced only when the free electron is scattered by the parent ion.

According to this view, the electron approaches the ion along one of two primary trajectories and shaping the electron trajectory is tantamount to controlling the HG process.[11] Several experimental observations, including the temporal coherence of the harmonic radiation,[12] structure in above threshold ionization (ATI) photoelectron energy spectra,[13] and the HG process itself[14] have been specifically attributed to these two primary trajectories or to interference between them.

Although semiclassical theories have been successful in describing the general characteristics of HG, they are unable to reflect the inherently quantum nature of processes such as ATI and HG. Since the semiclassical models rely on the acceleration of a free electron in the optical field and the subsequent recombination of that electron with an ion as the radiation production mechanism, they do not offer a physical process for producing the discrete photon and photoelectron energies observed experimentally. Previous theoretical treatments of HG have generally calculated the dipole moment for the electron oscillating in a classical optical field and have associated the terms of the Fourier transform of the dipole moment with the odd harmonics. This approach leaves unresolved the mechanism(s) by which the harmonics are generated and does not account for the quantized nonzero energies of the photoelectrons produced by ATI. In order to conserve momentum and energy, the semiclassical picture requires that the electron be born in the continuum with zero energy. Experiments, on the other hand, have demonstrated that ATI photoelectron energies are not continuous but rather take on discrete values.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

2.2. *The KFR approximation*

In his seminal paper published in 1935, D. M. Volkov[15] presented the solutions to the time-dependent Schrödinger's equation for an electron interacting with a classical optical field. Known as (classical) Volkov states, these solutions are at the heart of KFR theory[16–18] which provides a non-perturbative, semiclassical model for the ionization of an atom in an intense optical field. Semiclassical in the sense that the optical field is not quantized, KFR theory is based on the dipole approximation and the assumption that the outgoing wave in electron–photon scattering can be represented by a classical Volkov state,[15] a linear combination of plane waves whose phases are governed by the classical vector potential $\mathbf{A}$. A more detailed description of KFR theory and its application to ATI, the photoionization of atoms in strong fields, can be found in Refs. 16–20. In providing a nonperturbative framework for describing atomic ionization in intense optical fields, the KFR approximation has been successfully applied to describing the general characteristics of the photoelectron spectra resulting from ATI of rare gas atoms in circularly or elliptically polarized fields.[20] However, it suffers from several drawbacks that limit its applicability to a broader range of intense field phenomena. Because classical Volkov states are not energy eigenstates, neither the energy nor momentum of the

electrons are well-defined. As noted by Freeman and Bucksbaum,[20] "They [classical Volkov states] are not eigenstates of the mechanical momentum, nor the energy". Consequently, energy and momentum conservation does not follow directly from the theory.

2.3. *Quantum Volkov states*

Recently, we proposed an alternative framework for describing the interaction of an atom or electron with an intense optical field.[21–23] In this picture, electrons and the photon field form quantized field (*quantum*) Volkov states, stationary solutions of the time-*independent* Schrödinger's equation first derived by D.-S. Guo *et al.*[24,25] and applied to ATI. Since the quantum Volkov states are stationary, energy conservation follows directly from the energy eigenvalues of the e^--photon field system. Also, momentum is well-defined and a mechanism for the exchange of energy between the fundamental and harmonic optical fields becomes available — namely, transitions between the quantum Volkov states. It is important to recognize that the interaction between the electron and fundamental and secondary optical fields through the quantum Volkov state results in the exchange of energy in units of the fundamental photon energy, $\hbar\omega$. In other words, the "currency of exchange" is the photon energy. The electron and photon field can be viewed as forming a complex that is the vehicle by which energy is absorbed from, or delivered to, the electron and the photon field.

Quantum Volkov states may be regarded as the electron-optical field analog of bound states of an atom. Indeed, comparison of the theory of HG reported in Ref. 21 with experimental results repeatedly demonstrates that the Volkov $\rightarrow$ Volkov state transition matrix element is capable of reproducing all of the experimentally-observed characteristics of harmonic radiation, including spectra and polarization characteristics.[22,23] In retrospect, this is not surprising. Atomic and molecular spectroscopy in the past century was undergirded by the quantum mechanical notion of optical transitions between stationary states of the electron-atom (or molecule) system. We might expect to find, therefore, as optical intensities move inexorably higher, the same principle at work between the optical field and an electron. Rather than serving solely as an external agent in accelerating an electron, the optical field is a vital participant in stimulating the formation of quantum Volkov states and contributing to the composition of the e^--photon transient "complex".

2.4. *Theoretical development of the Volkov $\rightarrow$ Volkov transition matrix element*

2.4.1. *Brief review*

A detailed derivation of the transition matrix element for harmonic generation has been published recently[21,23] and only the salient points will be reviewed here. If the Coulombic interaction between an electron and the ion core is ignored, the matrix

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

element for *above threshold ionization* can be expressed as[24]

$$T_{\rm fi} = \sum_{\mathbf{P}_1, n} \langle \Phi_{\rm f}; n | \Psi_\mu \rangle \frac{i\alpha}{E - E_\mu + i\alpha} \langle \Psi_\mu | V_1 | \Phi_{\rm i}; \ell \rangle \,, \tag{1}$$

where $\mathbf{P}_1$ is the momentum of an ATI photoelectron, $\Phi_{\rm i}$ is a bound atomic state that is decoupled from the photon state $|\ell\rangle$ of the fundamental optical field, $\Phi_{\rm f}$ is the final state that is also decoupled from the fundamental field, α is the spectral linewidth of the fundamental radiation, and the electron-optical field interaction is given by

$$V_1 = -\frac{e}{m} \mathbf{A}(\mathbf{k} \cdot \mathbf{r}) \cdot (-i\boldsymbol{\nabla}) + \frac{e^2 A^2 (\mathbf{k} \cdot \mathbf{r})}{2m} \,, \tag{2}$$

in which the vector potential is related to the phase of the photon field, Θ, by the expression

$$\mathbf{A}(\mathbf{k} \cdot \mathbf{r}) = g\big(\hat{\varepsilon} e^{i\mathbf{k}\cdot\mathbf{r}} e^{i\Theta} \hat{a} + \hat{\varepsilon}^* e^{-i\mathbf{k}\cdot\mathbf{r}} e^{-i\Theta} \hat{a}^\dagger\big) \,. \tag{3}$$

In Eq. (3), g is the normalization factor arising from quantization of the optical field and $\hat{\varepsilon}$ is the polarization vector. The crucial feature of Eq. (1) is the introduction of quantized field (quantum) Volkov states Ψ_μ (Ref. 25) as intermediate states in the overall process where the wavefunctions

$$|\Psi_\mu\rangle = |\mathbf{P}_s, n_s\rangle = V_{\rm e}^{-1/2} e^{i(\mathbf{P}_s + z\mathbf{k})\cdot\mathbf{r}} e^{-iN_a \mathbf{k}\cdot\mathbf{r}} e^{-F_s} |n_s\rangle \tag{4}$$

are the solutions of the equation

$$(K + V_1)|\Psi_\mu\rangle = E_\mu |\Psi_\mu\rangle = E_\mu |\mathbf{P}_1, n_1; 0\rangle \,, \tag{5}$$

in which $K = (-i\boldsymbol{\nabla})^2/2m + N_a\omega$ is the non-interacting Hamiltonian for the electron and optical field, m is the electron rest mass, N_a is the number operator for the photon field, $V_{\rm e}$ is the normalization volume for the electron, z is the ponderomotive energy ($U_p = e^2\mathcal{E}^2(4m\omega^2)^{-1}$) normalized to the photon energy, ω is the frequency of the fundamental field, $\mathcal{E}$ is the classical amplitude of the electric field, the subscript s in Eq. (4) denotes a specific Volkov state ($s = 1, 2, \ldots$), and the coupling between the optical field and the electron is represented by the operator $\exp(-F_s)$, where

$$F_s = \frac{eg}{m\omega} (\mathbf{P}_s \cdot \hat{\varepsilon} e^{i\Theta_s} \hat{a} - \mathbf{P}_s \cdot \hat{\varepsilon}^* e^{-i\Theta_s} \hat{a}^\dagger)$$

$$+ \frac{e^2 g^2 \cos\xi}{4m\omega} \left(e^{-i2\Theta_s} \hat{a}^{\dagger 2} - e^{i2\Theta_s} \hat{a}^2 \right) . \tag{6}$$

In Eqs. (1)–(6), $\hbar \equiv {\rm c} \equiv 1$, and ξ expresses the degree of polarization of the optical field. The latter is related to the ellipticity μ by the relation

$$\mu = \tan \frac{\xi}{2} \,. \tag{7}$$

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

Before proceeding, it is important to remember that the Ψ_μ are the (stationary) solutions of the time-independent Schrödinger's equation (Eq. (5)) and that the eigenenergies of the quantum Volkov states are given by

$$E_\mu = \frac{|\mathbf{P}_1|^2}{2m} + n\omega + z\omega\,. \tag{8}$$

Notice that the energy eigenvalues depend on both the ponderomotive potential and the ATI electron energy but, more importantly, differ by integral multiples of the fundamental field photon energy. This characteristic of quantum Volkov states is matched by the inherently quantized nature of ATI, LAPE, and LAAD photoelectron energy spectra, for example, in which peaks are separated by multiples of the photon energy. A similar statement is true of HG radiation spectra. This correlation is a direct reflection of the centrality of the quantum Volkov state to all of these intense-field processes.

Although Eq. (1) provides an accurate description of the ATI process, it offers no means for generating a secondary radiation (harmonic) field and is, therefore, not suitable for representing HG. Since the harmonic intensities are generally considerably smaller than that for the fundamental, a secondary radiation field can readily be introduced by invoking time-independent perturbation theory. Doing so results in the following expression for the perturbed Volkov state, Ψ'_μ,

$$\Psi'_\mu = -\sum_\nu \frac{|\Psi_\nu\rangle\langle\Psi_\nu|V'|\Psi_\mu\rangle}{E_\nu - E_\mu}\,, \tag{9}$$

in which $|\Psi_\nu\rangle$ are the quantum Volkov state solutions for the equation:

$$(K + V_2)|\Psi_\nu\rangle = E_\nu|\Psi_\nu\rangle = E_\nu|\mathbf{P}_2, n_2; 1\rangle\,, \tag{10}$$

and whose eigenenergies are given by

$$E_\nu = E(|\mathbf{P}_2, n_2; 1\rangle) = \frac{|\mathbf{P}_2|^2}{2m} + n_2\omega + \frac{e^2g^2n_2}{m} + \omega'\,. \tag{11}$$

Notice in Eqs. (10) and (11) that one photon is now generated in the secondary field and the harmonic frequency is denoted ω'. Upon substituting $\Psi_\mu + \Psi'_\mu$ for Ψ_μ in Eq. (1), the transition matrix for the dominant term contributing to harmonic generation is denoted $T_{\mathrm{fi}}^{\mathrm{HG}}$ and is expressed as[21,23]

$$\begin{aligned} T_{\mathrm{fi}}^{\mathrm{HG}} = &\sum_{(\mathbf{P}_1, n_1)(\mathbf{P}_2, n_2)} \langle\Phi_{\mathrm{f}}; m; 1|\mathbf{P}_2, n_2; 1\rangle \\ &\times \langle\mathbf{P}_2, n_2; 1|V'|\mathbf{P}_1, n_1; 0\rangle\langle\mathbf{P}_1, n_1; 0|V_1|\Phi_{\mathrm{i}}; \ell; 0\rangle\,, \end{aligned} \tag{12}$$

where primes denote quantities associated with the secondary radiation field, the subscripts 1 and 2 represent parameters prior to and following the Volkov $\rightarrow$ Volkov state transition, respectively, V_2 differs from V_1 only by the shift in the phase of the fundamental field, $\Delta\Theta = \Theta_2 - \Theta_1$, that occurs as a result of the harmonic

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

generation (electron–photon field scattering) process, and the interaction term V' is given by

$$V' = -\frac{e}{m}\mathbf{A}'(\mathbf{k}'\cdot\mathbf{r})\cdot(-i\boldsymbol{\nabla}) + \frac{e^2}{m}\mathbf{A}(\mathbf{k}\cdot\mathbf{r})\cdot\mathbf{A}'(\mathbf{k}'\cdot\mathbf{r})\,. \tag{13}$$

The second term on the right-hand side of Eq. (13) is of particular importance because it describes the interaction between the fundamental and harmonic radiation fields. Generally, when the electron-field interaction is described by the minimum coupling expression, $(\mathbf{P}-e\mathbf{A}-e\mathbf{A}')^2/2m = \cdots-(e/m)\mathbf{P}\cdot\mathbf{A}-(e/m)\mathbf{P}\cdot\mathbf{A}'+(e^2/2m)\mathbf{A}\cdot\mathbf{A}'$, the last term is ignored. However, the intensities of both the fundamental and harmonic fields that are now available experimentally argue for retention of the $\mathbf{A}\cdot\mathbf{A}'$ term. It is this second order term that represents the exchange of energy between the fundamental and secondary optical fields by means of quantum Volkov states.

The recurring theme of this theory is that the middle term of Eq. (12), denoted T^{V}, is the matrix element that describes a new mechanism for the generation of harmonic radiation — optical transitions between quantum Volkov states. Notice in Eq. (12) that T^{V} is responsible for generating a photon in the harmonic field. There is no classical analog to this mechanism and it represents a clear departure from the currently-accepted picture of harmonic generation in which harmonic radiation is produced by the recombination of an electron with the parent ion. Conserving momentum and energy in the harmonic generation process requires that Φ_{f} in Eq. (12) be the initial, bound state of the atom (or molecule). This constraint requires that $\mathbf{P}_2 = \mathbf{P}_1 \equiv \mathbf{P}$. Therefore, QED theory predicts that the electron momentum $\mathbf{P}$ (determined by the number of photons absorbed during ATI) is not changed by Volkov $\rightarrow$ Volkov state transitions. Secondly, it is the quantum Volkov state that is responsible for converting fundamental photons into harmonic photons. To summarize, if the interaction of an electron with an optical field is expressed in terms of quantum Volkov states, then introducing a secondary photon field leads directly to the result that the harmonic (secondary) photons are produced by transitions between the quantum Volkov states. This is the single most significant result of quantizing the fundamental photon field.

Evaluating $T_{\mathrm{fi}}^{\mathrm{HG}}$ requires summing over $\mathbf{P}$, n_1 and n_2, which is equivalent to accounting for the contributions of all ATI processes. For a specified harmonic $q \equiv n_1 - n_2$, the harmonic generation transition matrix can be written

$$\begin{aligned} T_q^{\mathrm{HG}} = e^{iq\left(\frac{\Delta\Theta}{2}+\phi\right)} \sum_j \left(\frac{g'\omega^2}{2\mathcal{E}}\right)(z-j)\sqrt{2m^3\omega^5(j-z-E_{\mathrm{b}})} \\ \times \int_0^{2\pi} d\psi \int_0^{\pi} \sin\theta\, d\theta |\Im_{-j}(\zeta_j,\eta,\phi)|^2 |\Phi_{\mathrm{i}}(\mathbf{P}_j + z\mathbf{k} - j\mathbf{k})|^2 \\ \times \left\{-\frac{2e\mathcal{E}}{m\omega^2}\mathbf{P}_j\cdot\hat{\varepsilon}'^{*}\Im_{-q}\left(x_j, y, -\phi-\frac{\pi}{4}\right)\right. \end{aligned}$$

$$+\frac{2e^2\mathcal{E}^2}{m\omega^3}e^{-i\left(\frac{\Delta\Theta}{2}+\phi\right)}\Im_{-q+1}\left(x_j,y,-\phi-\frac{\pi}{4}\right)\hat{\varepsilon}\cdot\hat{\varepsilon}'^{*}$$

$$+\frac{2e^2\mathcal{E}^2}{m\omega^3}e^{i\left(\frac{\Delta\Theta}{2}+\phi\right)}\Im_{-q-1}\left(x_j,y,-\phi-\frac{\pi}{4}\right)\hat{\varepsilon}^{*}\cdot\hat{\varepsilon}'^{*}\Bigg\}\,, \tag{14}$$

where j is an integer representing the number of fundamental photons absorbed in the ATI process, E_b is the ionization potential of the atom or molecule of interest, $\Phi_{\rm i}(\mathbf{P}_j+z\mathbf{k}-j\mathbf{k})$ is the momentum-space representation (Fourier transform) of the bound atomic state wavefunction, $\Im$ is the generalized Bessel function,[23] and

$$x=\left|\frac{4e\mathcal{E}}{m\omega^2}\mathbf{P}_1\cdot\hat{\varepsilon}\right|\sin\frac{\Delta\Theta}{2}\,, \tag{15}$$

$$y=-z\cos\xi\sin\Delta\Theta\,, \tag{16}$$

and

$$\phi=\mathrm{Arg}(\mathbf{P}_1\cdot\hat{\varepsilon})\,. \tag{17}$$

The harmonic generation rate for the qth harmonic is then given by

$$W_q(\Delta\Theta)=\frac{V'_\gamma}{(2\pi)^3}\int d\Omega_{\mathbf{k}'}2\pi|T_q^{\rm HG}|^2=\frac{V'_\gamma}{2\pi^2}\int_0^{2\pi}d(\Delta\Theta)|T_q^{\rm HG}|^2 \tag{18}$$

where V'_γ is the photon normalization volume, and $\mathbf{k}'=q\mathbf{k}$ denotes the propagation vector for the harmonic optical field.

With the analytic expressions of Eqs. (14) and (18), one is able to calculate the harmonic spectrum or characteristics (polarization behavior, emission spectral lineshape) of a specific harmonic. Each is expressed in terms of the probability associated with transitions between pairs of quantum Volkov states. The requirement that the fundamental optical field experiences a phase shift as a result of Volkov $\rightarrow$ Volkov state transitions is a key prediction of this theory. It is well known that scattering processes are associated with a phase change. Electron scattering from a potential, for example, gives rise to a phase shift that determines the scattering cross-section.[26] Other contexts in which a phase shift is an integral part of the scattering process include the tunneling time asymmetry observed when electrons traverse semiconductor heterostructures[27,28] and the reflective properties of optical multilayer structures.[29]

Before proceeding to compare the predictions of this theory with experimental results, it should be mentioned that describing harmonic generation by formal scattering theory yields an expression that is essentially identical to the transition matrix of Eq. (12). A brief derivation of the HG scattering matrix can be found in the Appendix.

3. Comparison Of Theory With Experiment

Comparisons of this quantum theory of HG with semiclassical theory and experimental studies of harmonic spectra and polarization measurements available in the

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

literature have been presented previously in Refs. 21–23. Further theoretical results will be given here. In addition to harmonic spectra and the polarization characteristics of individual harmonics, calculations of harmonic spectral profiles will be presented and compared with the experimental results available in the literature.

3.1. *Harmonic spectra*

Figure 1, reprinted here for convenience from Ref. 23, illustrates harmonic spectra calculated from Eqs. (14) and (18) and compares the results with the semiclassical theory of Kan *et al.*[30] All of the spectra shown correspond to a fundamental field wavelength and intensity of 750 nm and 2×10^{15} W $\cdot$ cm^{-2}, respectively. Calculated from the formalism introduced in Ref. 31, the semiclassical spectra of Fig. 1 clearly extend to shorter wavelengths as the laser pulse width is reduced from 200 fs to 5 fs. Although the QED calculations shown in the figure do not depend explicitly on the laser pulse width, spectra are presented for several sets of j values. Notice that the quantum spectrum incorporating only the lowest two photoelectron orders ($j = 79, 80$) is quite similar to the semiclassical spectrum calculated for a 200 fs laser pulse. The slopes of the QED and semiclassical spectra in the plateau are virtually identical and, although the quantum spectrum falls off more quickly in intensity at high photon energies than does the 200 fs spectrum, both cut off in the

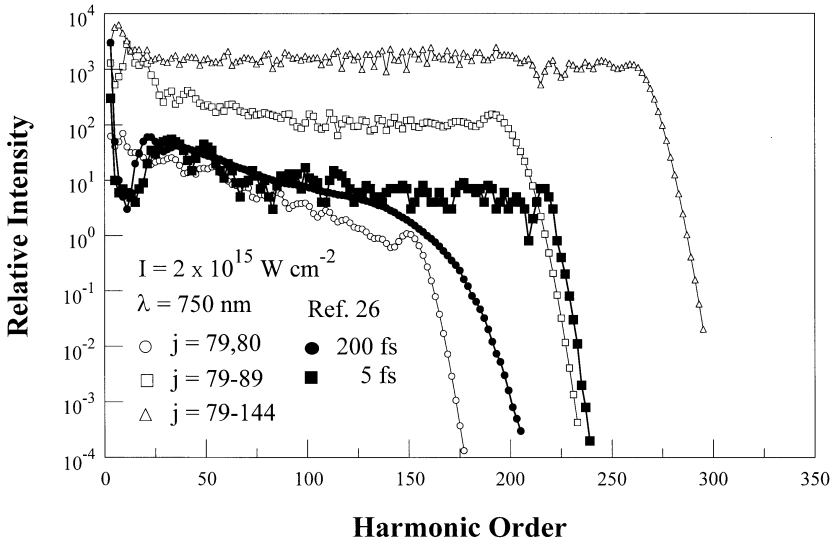


Fig. 1. Harmonic spectra calculated for a fundamental wavelength and intensity of 750 nm and 2×10^{15} W $\cdot$ cm^{-2}, respectively. The semiclassical calculations of Kan *et al.* (Ref. 30), based on the theory of Ref. 31, assume laser pulse widths of 5 fs (■) and 200 fs (●). The QED theoretical spectra, calculated from Eqs. (14) and (18), are not explicitly dependent upon laser pulse width but results are shown for several sets of j values. For the laser intensity and wavelength noted above, convergence of Eq. (14) in the text occurs when 65 photoelectron orders ($j = 79$–144) are included in the calculations.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

same region ($135 \lesssim q \lesssim 150$). Similar comments can be made regarding the 5 fs spectrum of Ref. 30 when compared with the spectrum calculated from the present theory if the lowest eleven photoelectron orders ($j = 79$–89) are included. Again, the plateau and cutoff portions of both spectra closely resemble one another, although the QED spectrum suggests that the relative harmonic intensity is roughly an order of magnitude higher than that predicted by semiclassical theory. The same trends are observed when comparing QED theory with experimental spectra.[23]

As was discussed in detail in Ref. 23, these results are consistent with the role of the photoelectron order in the QED theory. Briefly, the physical effect of reducing the laser pulse width for a given fundamental intensity is to effectively trap or "freeze" photoelectrons of increasingly higher orders. That is, reductions in pulse width utilize higher energy photoelectrons more effectively because the peak optical field intensity is reached before the electron escapes from its parent ion. Including more energetic photoelectrons into the calculations extends the harmonic response to shorter wavelengths because the coupling of a photoelectron to the optical field increases as the square root of the electron energy. Stated another way, although the QED theory described here and in Refs. 21–23 is time-independent, it is, nevertheless, linked with the temporal domain through the energies of the photoelectrons.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

3.2. *Polarization data: Dependence of harmonic ellipticity on the fundamental field ellipticity*

In Refs. 22 and 23, the predictions of quantum theory regarding the polarization characteristics of individual harmonics were compared with the available experimental data. In this section, we provide further comparisons with the data reported by Antoine and co-workers.[32]

It is well-known that the behavior of the polarization of harmonics in response to an elliptically-polarized fundamental provides a demanding test of any theory of harmonic generation. In Refs. 22 and 23, it was demonstrated that QED theory describes the experimental data in the literature remarkably well. Measurements of harmonic polarization characteristics have thus far taken on three forms. All involve varying the ellipticity of the fundamental optical field while recording either: (1) the relative harmonic yield, (2) the ellipticity of the harmonic, or (3) the angle by which the major axis of the harmonic polarization ellipse is rotated relative to that of the fundamental. In the discussion to follow, measurements of the dependence of harmonic ellipticity on the fundamental field ellipticity will be compared with QED theoretical calculations.

Figure 2 shows the calculated dependence of the ellipticity of the 29th harmonic of 790 nm in Ne on the ellipticity of the fundamental field for a fundamental intensity of 4×10^{14} W $\cdot$ cm^{-2}. The QED results, shown for two sets of j values ($j = 29, 30$ ($\circ$), and $j = 29$–31($\square$) where $j_{\min}$, the minimum number of fundamental photons required to ionize the atom for a given optical field intensity, is 29 in this case), are denoted by the open symbols and the measured values and semiclassical

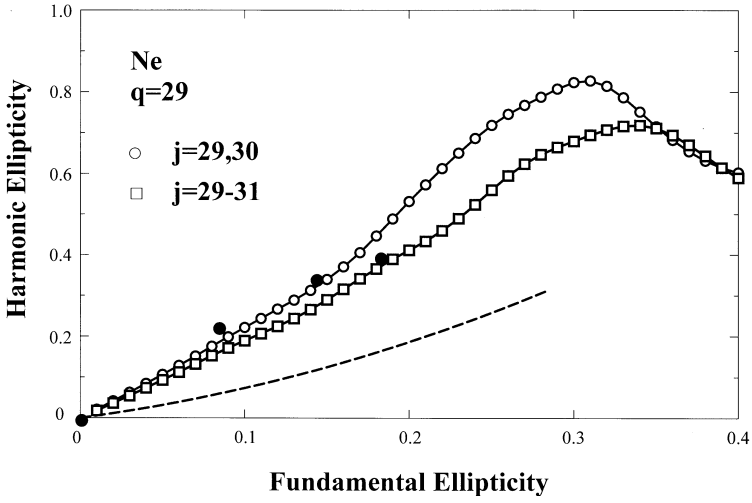


Fig. 2. Calculated dependence of the ellipticity of the 29th harmonic of 790 nm in Ne on the ellipticity of the fundamental field for an intensity of 4×10^{14} W·cm^{-2}. Results are shown for two sets of j values: (○) $j = j_{\min}$, $j_{\min} + 1 = 29, 30$, and (□) $j = 29$–31. The theoretical (ellipticity upper bound) and experimental results of Ref. 32 are represented by the dashed curve and solid circles (●), respectively.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

calculations of Ref. 32 are also given for the sake of comparison. The experimental measurements are in agreement with the present theory whereas the upper bound for the harmonic ellipticity, calculated from semiclassical theory, is more than a factor of two smaller than the experimental values. Notice, too, that QED theory predicts a rollover in the harmonic ellipticity for fundamental ellipticities above ~ 0.3. From the results of Fig. 2 as well as the comparisons of QED and semiclassical theory for HG in Refs. 22, 23, and 32, it is clear that semiclassical theory does not predict the rapid increase in harmonic ellipticity with rising fundamental ellipticity that is observed experimentally in Ne. Although the discrepancy between semiclassical theory and experiment is most severe for Ne (Refs. 22 and 23), it is also obvious in HG calculations for Ar.

A similar comparison between the semiclassical and quantum theoretical predictions of harmonic ellipticity is given in Fig. 3 for the 17th and 23rd harmonics of 790 nm in Ar. For all of the results shown, the fundamental intensity was assumed to be 2×10^{14} W · cm^{-2}. Once again, the values calculated from QED theory are presented for two sets of photoelectron orders (○: $j = 18, 19$; □: $j = 18$–20) whereas the semiclassical results (harmonic ellipticity upper bound) of Ref. 32 are illustrated by the dashed curves. As discussed previously,[23] the harmonic ellipticity calculations are not as sensitive to the choice of j values as are the calculations of rotation angle. This statement is particularly true for Ne. For both the Ar $q = 17$ and 23 data of Fig. 3, the QED theory predicts harmonic ellipticities that are *larger* than the observed values whereas the semiclassical theory underestimates the harmonic

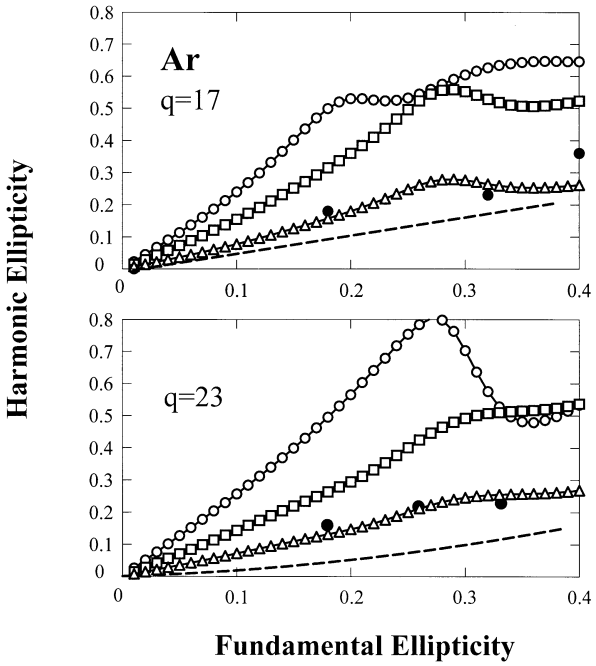


Fig. 3. Variation of the ellipticity of the 17th and 23rd harmonics of 790 nm in Ar with the ellipticity of the fundamental optical field. The fundamental intensity is assumed to be 2×10^{14} W·cm^{-2} for both sets of calculations. The results for the present theory are shown for two sets of photoelectron orders (○: $j = 18, 19$, and □: j = 18–20) and are compared with semiclassical calculations (dashed curve) and the measurements (●) of Ref. 32. The open triangles (△) represent the QED theoretical values for j = 18–20 (□) scaled down in magnitude by a factor of two.

ellipticity. Scaling the j = 18–20 curves of Fig. 3 down by a factor of two (△) yields reasonable agreement with the magnitude and functional dependence of the experimental data.

3.3. *Spectral lineshapes*

Calculations of the spectral profiles for specific harmonics have also been carried out recently and several results are reported here. For a given harmonic q, the lineshape $g(\omega')$ can be determined from the expression:

$$g(\omega') = \int \frac{\int_0^{2\pi} |T_q|^2 d\Theta}{(\omega' - q\omega)^2 + \alpha^2} d\omega\,, \tag{19}$$

where α again represents the linewidth of the fundamental, ω' is the harmonic frequency, and T_q is the harmonic generation transition matrix (Eq. (14)). Assuming the bandwidth of the fundamental to be 14 nm (Ref. 12), the spectral lineshape for the 7th harmonic of 790 nm in Ar for a fundamental intensity of 2×10^{14} W·cm^{-2} is illustrated in Fig. 4. In fact, several profiles are shown, each of which is associated

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

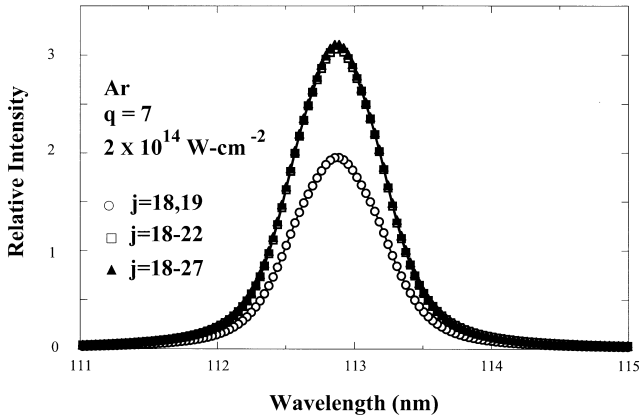


Fig. 4. Spectral profile for the 7th harmonic of 790 nm in Ar for a fundamental intensity and bandwidth of 2×10^{14} W $\cdot$ cm^{-2} and 14 nm, respectively. Results are shown for several groups of photoelectron orders: $j = 18, 19$ (○), $j = 18$–22 (□), and $j = 18$–27 (▲). Note the rapid convergence of the lineshape to a bandwidth (FWHM) of ~ 0.8 nm.

with a range of j values (○: $j = 18, 19$; □: $j = 18$–22; ▲: $j = 18$–27). The key point to be made is that, as expected and observed previously for harmonic spectra (Ref. 23), the lineshape function $g(\omega')$ converges as the number of photoelectron orders incorporated into the calculations is increased. Furthermore, convergence occurs quickly. The differences between the profiles calculated when including five ($j = 18$–22) and ten ($j = 18$–27) photoelectron orders in the calculations are negligible.

Few experimental measurements of harmonic spectral widths exist in the literature with which the predictions of quantum theory may be compared. However, Bellini *et al.*[12] have measured the temporal coherence of the 7th–21st harmonics of 790 nm in Ar by a clever interferometric technique. From the modulation depth (fringe visibility) of the interference fringe pattern produced by a pair of fundamental pulses separated by a variable time delay, they were able to infer harmonic bandwidths between 0.1 and 0.2 nm. These spectral widths correspond to coherence times of 20–40 fs which are consistent with the expected temporal width of the harmonic pulses.

Figure 5 shows the calculated lineshapes for the $q = 7$, 11, 15, and 23 harmonics of 790 nm in argon when the bandwidth and intensity of the fundamental radiation are taken to be 14 nm and 10^{14} W $\cdot$ cm^{-2}, respectively. Again, results are shown for two sets of j values (solid curve: $j = 14, 15$; dashed curve: $j = 14$–16) and it should be reiterated that these calculations represent only the single atom response — propagation effects have not been included. For the sake of comparison of the harmonic spectral profiles, Fig. 6 is a graph of the bandwidth (FWHM) of the four harmonics in Fig. 5 (and one additional harmonic, $q = 19$), shown as a function of harmonic order. Notice that the bandwidth falls monotonically, from $\sim$0.8 nm for

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

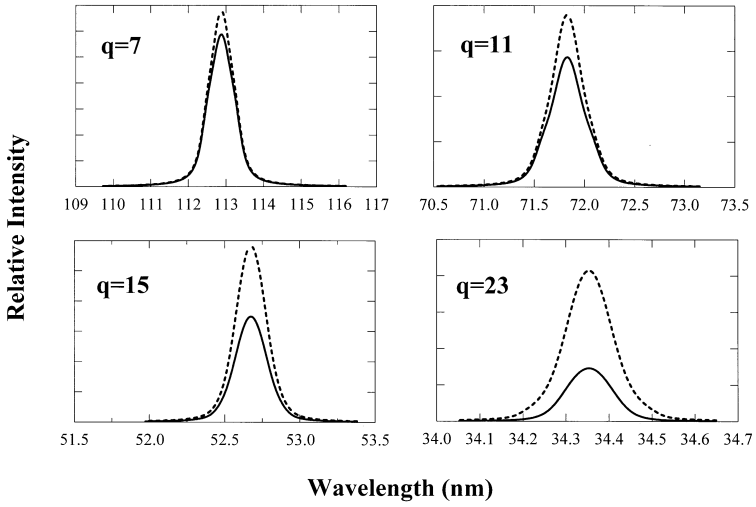


Fig. 5. Spectral lineshapes calculated from Eq. (19) for the 7th, 11th, 15th, and 23rd harmonics of 790 nm in Ar when the fundamental intensity is fixed at 10^{14} W $\cdot$ cm^{-2}. The bandwidth of the fundamental is taken to be 14 nm (Ref. 12) and profiles associated with two sets of j values (solid curve; $j = 14, 15$: dashed curve: $j = 14$–16) are shown.

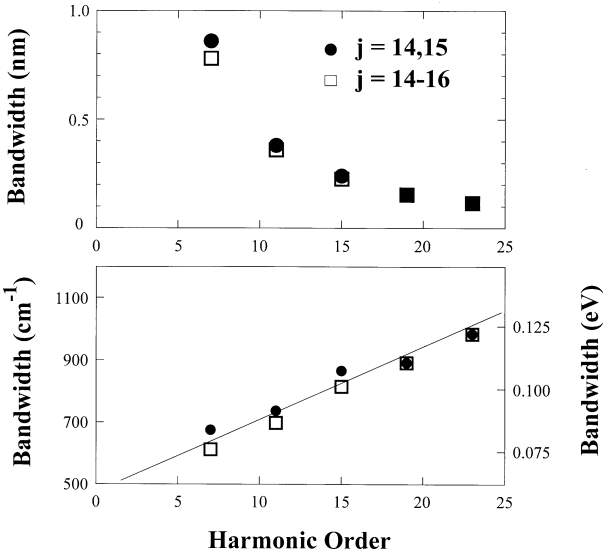


Fig. 6. Top: Dependence of the harmonic bandwidth (FWHM, nm) on the harmonic order, q, for a fundamental wavelength and intensity of 790 nm and 10^{14} W $\cdot$ cm^{-2}, respectively. These results were extracted from Fig. 5 and results for two sets of photoelectron orders ($j = j_{\min}, j_{\min} + 1$, and $j = j_{\min} - j_{\min} + 2$) are shown. Bottom: The same data but the FWHM is now expressed in cm^{-1} or eV. The solid line is the linear least-squares fit to the data.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

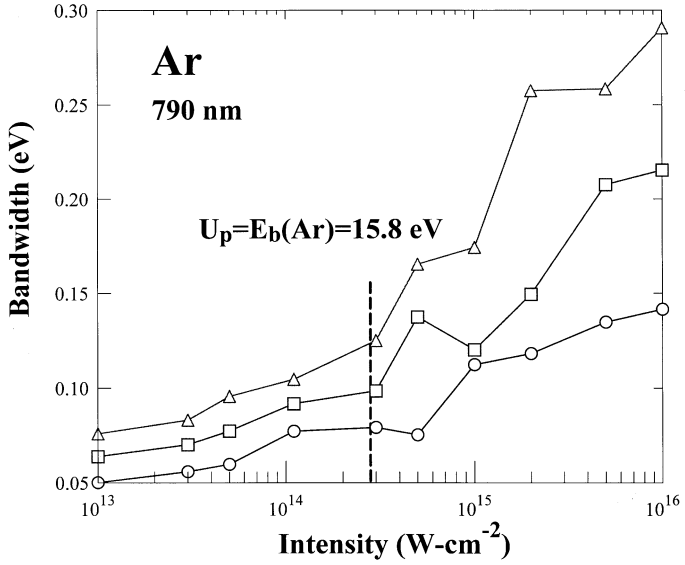


Fig. 7. Variation of the bandwidth (FWHM) of the 7th (○), 11th (□), and 15th (Δ) harmonics of 790 nm, generated in Ar, with the fundamental intensity over the range $10^{13} - 10^{16}$ W-cm^{-2}. The two lowest photoelectron orders are incorporated into the calculations and the bandwidth of the fundamental is assumed to be 14 nm.

$q = 7$ to ∼0.23 nm for $q = 15$ to ∼0.13 nm for $q = 23$. Expressed in terms of energy (Fig. 6, lower panel), the harmonic bandwidth is predicted to increase linearly with q, to a value of ∼120 meV (or ∼4.5 kT at 300 K) for $q = 23$. The slope of the data in the lower panel of Fig. 6 is ∼3 meV/harmonic. We note that the results of Fig. 6 are in general accord with the data of Bellini *et al.*[12] who, as one might recall, inferred bandwidths between 0.1 and 0.2 nm for the $q = 7$–23 harmonics of 790 nm in Ar. Although the bandwidth for $q = 7$(∼0.8 nm) is well outside the 0.1–0.2 nm range, the average of the bandwidths for $q = 11, 15, 19$, and 23 is 0.22 ± 0.11 nm.

It is also instructive to examine the dependence of harmonic bandwidth on the fundamental intensity. As illustrated in Fig. 7 for harmonic generation in Ar, increasing the fundamental ($\lambda = 790$ nm) intensity from 10^{13} W · cm^{-2} to 10^{16} W · cm^{-2} results in an almost four-fold increase in the FWHM of the emission profile for the 15th harmonic, for example. It is interesting to note that the harmonic bandwidth initially rises slowly until the intensity is reached ($I \backsimeq 2.7 \times 10^{14}$ W · cm^{-2}) at which U_p, the ponderomotive energy, becomes comparable to the ionization energy of Ar (∼15.8 eV). For still higher intensities, the harmonic linewidth increases sharply. Similar behavior is observed for the other two harmonics except that the slope of the curves for $I \gtrsim 3.10^{14}$ W-cm^{-2} increases with harmonic order. Thus, the calculations predict that the higher harmonics broaden more quickly with rising intensity than do the longer wavelength (lower q) harmonics.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

4. Summary and Conclusions

A quantum theory of high order harmonic generation has been developed in which transitions between quantum Volkov states is a new mechanism for producing harmonic radiation. In this QED picture of HG, the momentum and energy of the electron are well-defined and the discrete, nonzero energies of the photoelectrons produced by ATI are directly incorporated into the theory. Most importantly, the production of radiation at specific wavelengths takes on physical significance in this picture. Rather than arising from the Fourier spectrum of the dipole moment, the harmonics result from transitions between states separated by multiples of the fundamental photon energy. Integrating over the photoelectron momentum $\mathbf{P}$ causes the even harmonics to vanish. There is no classical analog to this process and the stationary (quantum Volkov) states at the heart of the theory can be regarded as the optical field analog of bound states of an atom.

This new picture of high order harmonic generation has proven to be remarkably successful in reproducing the results of experiments reported to date, several of which, to our knowledge, have not been matched by semiclassical theory. Harmonic spectra, a wide range of polarization data, and harmonic lineshapes are described well by the transition matrix element, for which an analytic expression has been derived.

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

Acknowledgment

The technical assistance of K. Collier is gratefully acknowledged. This work was supported by the U.S. Air Force Office of Scientific Research.

Appendix: Scattering Matrix Representation of Harmonic Generation

It will be shown here that the harmonic generation process can be described by formal scattering theory and, in so doing, the result is identical to that obtained by calculating the transition matrix. The scattering matrix S_{fi} can be written as

$$S_{\mathrm{fi}} = \langle \psi^- | \psi^+ \rangle \,, \tag{A.1}$$

where ψ^+ and ψ^-, the incoming and outgoing scattering wavefunctions, are given by:

$$\psi^+ = \frac{i\alpha}{E - K - V_1 - i\alpha} |\Phi_{\mathrm{i}}; \ell; 0\rangle \,, \tag{A.2}$$

and

$$\psi^- = \frac{-i\alpha}{E - K - V_2 - V' + i\alpha} |\Phi_{\mathrm{f}}; m; 1\rangle \,, \tag{A.3}$$

where K and V_1 are defined as before, and V_2 once again differs from V_1 only by the phase shift $\Delta\Theta = \Theta_2 - \Theta_1$ which is a result of the electron–photon field scattering

process. That is:

$$V_s = -\frac{e}{m}\mathbf{P}\cdot\mathbf{A}(\Theta_s) + \frac{e^2}{2m}\mathbf{A}^2(\Theta_s), \qquad s = 1, 2\,. \tag{A.4}$$

In this formalism, the Coulombic interaction has been neglected in the strong field approximation, and the laser-atom interaction is described by initial and final states ($|\Phi_\mathrm{i}; \ell; 0\rangle$ and $|\Phi_\mathrm{f}; m; 1\rangle$, respectively) representing the laser field decoupled from the atom. Notice that the photon number for the laser field is changed by the scattering process from ℓ to m and one photon is now generated in the secondary (harmonic) field. Also, α represents the linewidth of the laser field. Consequently, one finds that

$$S_\mathrm{fi} = \langle\Phi_\mathrm{f}; m; 1\left|\frac{i\alpha}{E - K - V_2 - V' + i\alpha}\cdot\frac{i\alpha}{E - K - V_1 + i\alpha}\right|\Phi_\mathrm{i}; \ell; 0\rangle\,. \tag{A.5}$$

In order to evaluate the Green's operators, the identity

$$|\Psi_n\rangle\langle\Psi_n| = |\Psi'_n\rangle\langle\Psi'_n| = 1 \tag{A.6}$$

is invoked where

$$(K + V_1)|\Psi_n\rangle = E_n|\Psi_n\rangle \tag{A.7}$$

and

$$|\Psi_n\rangle = |\mathbf{P}_1, n_1; 0\rangle\,. \tag{A.8}$$

Similarly,

$$(K + V_2 + V')|\Psi'_n\rangle = E'_n|\Psi'_n\rangle \tag{A.9}$$

where

$$|\Psi'_n\rangle = |\mathbf{P}_2, n_2; 1\rangle\,, \tag{A.10}$$

the Ψ_n and Ψ'_n are quantum Volkov states, and primes again denote parameters associated with the harmonic field. As a result, the scattering matrix element can be expressed as:

$$S_\mathrm{fi} = \sum_{n_1, n_2}\langle\Phi_\mathrm{f}; m; 1|\frac{i\alpha}{E - E'_n + i\alpha}|\Psi'_n\rangle$$

$$\times\,\langle\Psi'_n|\Psi_n\rangle\langle\Psi_n|\frac{i\alpha}{E - E_n + i\alpha}|\Phi_\mathrm{i}; \ell; 0\rangle\,. \tag{A.11}$$

If $V' \ll V_1 \approx V_2$, then

$$S_\mathrm{fi} = \sum_{(\mathbf{P}_1, n_1)(\mathbf{P}_2, n_2)}\langle\Phi_\mathrm{f}; m; 1|\frac{i\alpha}{E - E'_n + i\alpha}|\mathbf{P}_2, n_2; 1\rangle$$

$$\times\langle\mathbf{P}_2, n_2; 1|V'|\mathbf{P}_1, n_1; 0\rangle\langle\mathbf{P}_1, n_1; 0|\frac{i\alpha}{E - E_n + i\alpha}|\Phi_\mathrm{i}; \ell; 0\rangle \tag{A.12}$$

which differs from the transition matrix ($T_\mathrm{fi}^\mathrm{HG}$, Eq. (12)) only by the linewidth factor. At resonance ($E_n \to E'_n$), one again finds that $\mathbf{P}_1 = \mathbf{P}_2$ (i.e., the electron

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

momentum is unchanged by the HG scattering process) and momentum conservation is enforced. Therefore, the relative intensity of the qth harmonic is given by:

$$I_q \propto |S_{\mathrm{fi}}|^2$$

$$= \left| \sum_{\substack{(\mathbf{P}_1,n_1)(\mathbf{P}_2,n_2) \\ n_1-n_2\equiv q}} \langle \Phi_{\mathrm{f}}; m; 1|\mathbf{P}_2, n_2; 1\rangle \langle \mathbf{P}_2, n_2; 1|V'|\mathbf{P}_1, n_1; 0\rangle \langle \mathbf{P}_1, n_1; 0|\Phi_{\mathrm{i}}; \ell; 0\rangle \right|^2 .$$

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.

References

1. A. McPherson, G. Gibson, H. Jara, U. Johann, T. S. Luk, I. McIntyre, K. Boyer and C. K. Rhodes, *J. Opt. Soc. Am.* **B4**, 595 (1987).
2. X. F. Li, A. L'Huillier, M. Ferray, L. A. Lompré and G. Mainfray, *Phys. Rev.* **A39**, 5751 (1989).
3. J. J. Macklin, J. D. Kmetec and C. L. Gordon III, *Phys. Rev. Lett.* **70**, 766 (1993).
4. See, for example, M. Schnürer, Ch. Spielmann, P. Wobrauschek, C. Streli, N. H. Burnett, C. Kan, K. Ferencz, R. Koppitsch, Z. Cheng, T. Brabec and F. Krausz, *Phys. Rev. Lett.* **80**, 3236 (1998).
5. See, for example, P. A. Norreys, M. Zepf, S. Moustaizis, A. P. Fews, J. Zhang, P. Lee, M. Bakarezos, C. N. Danson, A. Dyson, P. Gibbon, P. Loukakos, D. Neely, F. N. Walsh, J. S. Wark and A. E. Dangor, *Phys. Rev. Lett.* **76**, 1832 (1996).
6. A. Weingartshofer, J. K. Holmes, G. Caudle, E. M. Clarke and H. Krüger, *Phys. Rev. Lett.* **39**, 269 (1977).
7. J. M. Schins, P. Breger, P. Agostini, R. C. Constantinescu, H. G. Muller, G. Grillon, A. Antonetti and A. Mysyrowicz, *Phys. Rev. Lett.* **73**, 2180 (1994).
8. T. E. Glover, R. W. Schoenlein, A. H. Chin and C. V. Shank, *Phys. Rev. Lett.* **76**, 2468 (1996).
9. K. C. Kulander, K. J. Schafer and J. L. Krause, in *Super-Intense Laser-Atom Physics*, eds. B. Piraux, A. L'Huillier and K. Rzążewski, NATO ASI Ser. B, Vol. 316 (Plenum, New York, 1993).
10. P. B. Corkum, *Phys. Rev. Lett.* 71, 1994 (1993).
11. M. Lewenstein, P. Salières and A. L'Huillier, *Phys. Rev.* **A52**, 4747 (1995).
12. M. Bellini, C. Lyngå, A. Tozzi, M. B. Gaarde, T. W. Hänsch, A. L'Huillier and C.-G. Wahlström, *Phys. Rev. Lett.* **81**, 297 (1998).
13. G. G. Paulus, F. Zacher, H. Walther, A. Lohr, W. Becker and M. Kleber, *Phys. Rev. Lett.* **80**, 484 (1998).
14. G. van de Sand and J. M. Rost, *Phys. Rev. Lett.* 83, 524 (1999).
15. D. M. Volkov, *Z. Phys.* **94**, 250 (1935).
16. L. V. Keldysh, *Sov. Phys. JETP* **20**, 1307 (1965) [*Zh. Th.* **P.47**, 1945 (1964)].
17. F. H. M. Faisal, *J. Phys.* **B6**, L89 (1973).
18. H. R. Reiss, *Phys. Rev.* **A22**, 1786 (1980).
19. H. R. Reiss, *J. Phys.* **B20**, L79 (1987).
20. R. R. Freeman and P. H. Bucksbaum, *J. Phys.* **B24**, 325 (1991).
21. J. Gao, F. Shen and J. G. Eden, *Phys. Rev. Lett.* **81**, 1833 (1998).
22. J. Gao, F. Shen and J. G. Eden, *J. Phys.* **B32**, 4153 (1999).
23. J. Gao, F. Shen and J. G. Eden, *Phys. Rev.* **A61**, 043812 (2000).
24. D.-S. Guo , T. Åberg and B. Crasemann, *Phys. Rev.* **A40**, 4997 (1989).

25. D.-S. Guo and G. W. F. Drake, *J. Phys.* **A25**, 3383 (1992).
26. C. Cohen-Tannoudji, B. Diu and F. Laloë, *Quantum Mechanics*, Vol. II (John Wiley and Sons, New York, 1977), pp. 931–934.
27. See, for example, V. Gasparian, T. Christen and M. Büttiker, *Phys. Rev.* **A54**, 4022 (1996).
28. D. Dragoman, *IEEE J. Quantum Electron.* **35**, 1887 (1999) and references cited therein.
29. J.-M. Vigoureux and D. Van Labeke, *J. Mod. Opt.* **45**, 2409 (1998).
30. C. Kan, N. H. Burnett, C. E. Capjack and R. Rankin, *Phys. Rev. Lett.* **79**, 2971 (1997).
31. M. Lewenstein, Ph. Balcou, M. Yu. Ivanov, A. L'Huillier and P. B. Corkum, *Phys. Rev.* **A49**, 2117 (1994).
32. P. Antoine, B. Carré, A. L'Huillier and M. Lewenstein, *Phys. Rev.* **A55**, 1314 (1997).

Int. J. Mod. Phys. B 2000.14:889-907. Downloaded from www.worldscientific.com
by UNIVERSITY OF TORONTO on 02/01/15. For personal use only.